# Exam Questions

1. 

2. Which quantity has the same units as those for energy stored per unit volume? A

3. 

4. An object of mass 8.0 kg is falling vertically through the air. The drag force acting on the object is 60 N. What is the best estimate of the acceleration of the object? B

5. 

6. A body is held in translational equilibrium by three coplanar forces of magnitude 3 N, 4 N and 5 N. Which statements are true? I. All forces are perpendicular to each other II. The forces cannot act in the same direction III. The vector sum of the forces is equal to zero. II, III

7. 

8. An object of mass 1 kg is thrown downwards from a height of 20 m with an initial speed of 6 m/s. The object hits the ground at a speed of 20 m/s. Using $g = 10\,ms^{-2}$, what is the best estimate of the energy transferred from the object to the air? C

9. 

10. A car is driven from rest along a straight horizontal road. The car engine exerts a constant driving force. Friction and air resistance are negligible. How does the power developed by the engine change with the distance travelled? C

11. 

12. An object of mass $2m$ moving at velocity $3v$ collides with a stationary object of mass $4m$. The objects stick together after the collision. A

13. 

14. What is not an assumption of the kinetic model of an ideal gas? A

15. 

16. A gas has a density about 1000 times less than its liquid form. Given molecular diameter $d$, what is the best estimate of the average intermolecular distance in the gas? B

17. 

18. Containers X and Y at same temperature; X: 4 $m^3$, 100 Pa; Y: 6 $m^3$, 50 Pa. They are joined. Final pressure? B

19.

20. An object moves with simple harmonic motion. The acceleration of the object is: A

21.

22. A wave has frequency 500 Hz. Phase difference of $\frac{\pi}{3}$ between points 0.050 m apart. Wave speed? A

23.

24. Wavefronts travel from air to medium Q. What is refractive index of Q? A

25.

26. Pipe open at both ends vibrates in second harmonic mode. What is the phase difference between particles at P and Q? A

27.

28. A wire has free charge density $n$, carrier charge $q$. What additional value is needed to find current per unit area? A

29.

30. A cell of emf $E$ and zero internal resistance is in a circuit. Which is correct for loop WXYUW? A

31.

32. Given constant resistance, the relationship between resistivity $\rho$, radius $r$, and length $l$ is: A

33.

34. Power station generates 250 kW at 25 kV. Cable resistance 4$\Omega$. Power loss? B

35.

36. Power supply with internal resistance supplies direct current $I$ for time $t$. EMF is: A

37.

38. A current in a wire lies between the poles of a magnet. What is the direction of the electromagnetic force on the wire? The direction of the electromagnetic force on the wire is perpendicular to both the direction of the current and the magnetic field, as determined by the right-hand rule.

39.

40. Mass $M$ is suspended, and $m$ moves in horizontal circle radius $r$ at speed $v$. What is $M$? A

41.

42. An atom has four energy states. Shortest radiation wavelength emitted is A

43.

44. Proton-proton collision. Feynman diagram changes. Reaction? A

45.

46. Developments: I. Wave-particle duality II. Kinetic model III. Mass-energy equivalence Which represented paradigm shifts? A

47.

48. Nuclear fuel decreases by 8 mg/hour. Efficiency = 50%. Max power output? A

49.

50. A nuclear power station has an AC generator. What energy transfer takes place in the generator? C

51.

52. Planet's surface temperature 5$\times$ its moon. Same emissivity. Planet intensity = $I$. Moon's intensity? A

53.

54. Simple pendulum and mass-spring: same period initially. Both masses halved. What is $\frac{\mathrm{T}_{\text{pendulum}}}{\mathrm{T}_{\text{spring}}}$? A

55.

56. Two spectral lines differ by 0.59 nm, mean $\lambda = 590$ nm. Resolved in 4th order. Minimum grating lines illuminated? B

57.

58. Gratings X and Y, Y has more lines per metre. Which are true? I. Slits are further apart for X II. Angle between red and blue is greater in X III. X gives more visible orders A

59.

60. Two satellites W and X same mass. W is closer to planet. What is smaller for W compared to X? C

61.

62. P, S: same equipotential. Q, R: same but farther from planet. Most work done by gravity going from: A

63.

64. Graph of $E$ vs. $r$, shaded area X between $r_1$ and $r_2$. X represents A

65.

66. Why use high voltage, low current over long distances? A

67.

68. Power $P$ in resistance $R$ with direct current $I$. What is power in $\frac{R}{2}$ with rms current $2I$? B

69.

70. Rectangular coil rotates. Plane perpendicular to magnetic field direction ZZ . What rotation produces graph of emf vs time? A

71.

72. Capacitor $C$, initial charge $Q$, discharges through resistor $R$, voltage varies with time. Which is true? A

73.

74. Light hits metal, emits electrons. Intensity increased. Changes to emission rate and KE? A

75.

76. Nucleus diameter = 12 fm. What is nucleon number? C

77.

78. Photon wavelength $\lambda$. What are energy and momentum? A

79.

80. What did Rutherford-Geiger-Marsden experiment show? A